

SUBORNO DEB BAPPON

Saskatoon, SK S7N 0V1

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EDUCATION

University of Saskatchewan

Sep. 2023 - Jan. 2026

Master of Science in Computer Science - Grade: 89.5%

Saskatoon, Canada

- Relevant Coursework: Software Maintenance & Evolution, Advanced Deep Learning, Human-Centric Software Renovation, Empirical Software Engineering
- Thesis: Enhancing Knowledge Quality of Crowd-Sourced Developer Q&A Platforms through AI-Driven Software Solutions
- Advisors: Dr. Chanchal K. Roy, Dr. Kevin Schneider

Chittagong University of Engineering & Technology

Feb. 2017 - Sep. 2022

Bachelor of Science in Computer Science & Engineering - CGPA: 3.76/4.00 (Merit: 3rd among 128)

Chattogram, Bangladesh

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Artificial Intelligence, Machine Learning, Database Management Systems

SKILLS

Languages: Python, SQL, C++, C, HTML, CSS, JavaScript

Generative AI & Machine Learning: OpenAI Agents SDK, CrewAI, FastMCP, LangChain, scikit-learn, PyTorch, TensorFlow

Frameworks: FastAPI, Flask, Django

Tools & Platforms: Git, GitHub, GitLab, AWS, Azure, VS Code, Cursor, JupyterLab, Linux, Windows, macOS

WORK EXPERIENCE

Software Developer - AI

Feb. 2026 - Present

Siemens

Saskatoon, Canada

- Designing and developing Generative AI and Agentic AI solutions for Siemens' Electronic Design Automation (EDA) systems.

Graduate Research Assistant — Software Research Lab

Sep. 2023 - Dec. 2025

University of Saskatchewan

Saskatoon, Canada

- Developed LLM-powered tools using REST APIs, enhancing code comment quality by 30% and achieving 84.5% developer adoption for *Stack Overflow* answer refinement.
- Built a RAG-powered browser plugin leveraging multimodal LLMs for real-time technical debt detection and classification on *Stack Overflow*, achieving 90.7% accuracy.
- Reduced software prototyping time by 60% and documentation workload by 80% via multi-agent AI systems for automated software design and documentation.

Graduate Teaching Assistant — Computer Science

Sep. 2023 - Apr. 2025

University of Saskatchewan

Saskatoon, Canada

- Guided 30+ students through data-driven programming projects, improving project completion rates by 20% through hands-on mentorship and code review sessions.
- Designed and implemented structured grading rubrics that identified key learning gaps, contributing to a 15% improvement in average final exam performance.
- Mentored students on software documentation, version control (Git), and code quality standards, resulting in a $\approx 15\%$ increase in industry-ready project submissions.

Lecturer — Computer Science & Engineering

Feb. 2023 - Jul. 2023

Eastern University

Dhaka, Bangladesh

- Delivered lectures and labs on Python, databases, AI, and machine learning, integrating real-world projects that boosted student lab performance by 22%.
- Mentored undergraduates in software development best practices, improving project completion rates by 30%.
- Modernized course content to align 80% of curriculum with industry topics like data analytics, machine learning and software engineering, enhancing internship readiness.

PROJECTS

Neora: Multi-Agent System for Rapid Software Prototyping – Python, CrewAI, OpenAI/Gemini API, Gradio, Docker

- Developed an autonomous multi-agent system that converts plain-text requirements into Python modules and UI prototypes, streamlining the MVP development process.
- Engineered modular YAML pipelines enabling adaptive role-specific reasoning across LLMs such as GPT-5 and Gemini 2.5 Flash.

Repolish: Automated README File Generator – Python, CrewAI, OpenAI/Gemini API, Gradio, CSS

- Built an AI-powered tool to parse GitHub repositories and auto-generate professional README files, optimizing documentation process.
- Integrated Gradio-based LLM interface and automated PR workflows, reducing release cycles from hours to minutes.

DEBug.me: Personal Agentic AI Assistant – Python, RAG, OpenAI Embeddings, Gemini API, Chroma, Gradio

- Built a RAG-powered AI assistant using Chroma vector search and multi-API integration for accurate, context-grounded responses.
- Deployed a Gradio-based web app with real-time chat, conversation memory, and NTFY-integrated user notifications.

Farmona: Smart Crop Prediction & Agronomic Analysis – Python, Scikit-learn, Pandas, NumPy, Streamlit

- Built an ML pipeline with feature engineering and GridSearchCV optimization, achieving 99% accuracy in crop prediction.
- Deployed an interactive Streamlit app for real-time agronomic insights with reproducible deployment artifacts.

SELECTED PUBLICATIONS

- S. D. Bappon, S. Mondal, and B. Roy. "AUTOGENICS: Context-Aware Inline Comment Generation Using LLMs." In *Proc. IEEE SCAM*, pp. 24–35, 2024.
- S. Mondal, S. D. Bappon, and C. K. Roy. "Enhancing ChatGPT User Interaction through Consolidated Prompts for Issue Resolution." In *Proc. IEEE/ACM MSR*, pp. 222–226, 2024.

CERTIFICATIONS

ML with Python (IBM), Neural Networks & Deep Learning (DeepLearning.AI), NLP with TensorFlow (DeepLearning.AI), Problem Solving (Intermediate) (HackerRank)

HONORS & AWARDS

Best Research Award, CMPT 885: Human-Centric Software Renovation, University of Saskatchewan (Spring 2024)

75th Anniversary Recruitment Scholarship, CGPS, University of Saskatchewan (2023 - 2025)

Dean's Award, Faculty of Electrical & Computer Engineering, Chittagong University of Engineering & Technology (Sep. 2022)

EXTRACURRICULAR ACTIVITIES

Vice President Internal, Computer Science Graduate Council, University of Saskatchewan (Sep. 2024 – Oct. 2025)

Mentor, Eastern University Computing Club (May 2023 – Jul. 2023)

REFERENCES

Available upon request.